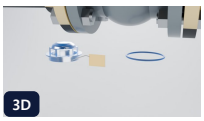
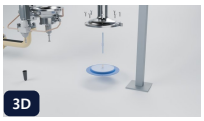
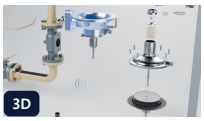
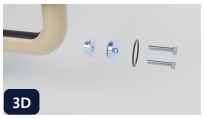
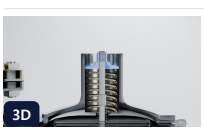
Fisher™ 299H regulator (with optional 67CF) — Step-by-step

One page to follow along with the ② disassembly & assembly guide video — the ▶ time on the right opens that part of the video.

② Disassembly guide
English

Work by qualified personnel following the manufacturer's manual · KFNGS training material — not an Emerson publication
Work the manual does not describe: do not proceed until the manufacturer's procedure is confirmed
torques from the manual only · tool sizes from KFNGS service records — verify the actual fastener

Photo	Step · what to do / see	Tools · torque	★ Caution · check	Video
 3D	**Before you start — isolation and pressure release** Isolate with the inlet and outlet block valves and release downstream, then inlet pressure. Confirm zero on the inlet, loading and outlet gauges (p. 9).	—	First confirm the vent path is safe.	▶ 0:08
 3D	**①② Pilot cap · wire seal · closing cap** Remove the pilot cap (29), loosen the locknut (35) and turn the adjusting screw (36) counterclockwise to ease spring compression (p. 11). Cut the wire seal and remove the closing cap (p. 9).	—	Keep the warning tag — refit the wire seal and tag after maintenance (p. 10).	▶ 1:13
 3D	**③ Adjusting nuts · spring seat · closing spring** Carefully remove the 2 adjusting nuts (5), then the spring seat (4) and closing spring (6) (p. 9).	—	★ The seat is under spring force · do not turn the diaphragm post (10) (pp. 9–10).	▶ 1:42
 3D	**④ Upper casing** Removing the upper casing with the loading tube (22) connected: do not proceed until the manufacturer's procedure is confirmed. Then follow p. 9 — remove the 8 hex cap screws (23) and lift off the upper casing.	**Assembly torque: 13–17 N·m, crisscross (p. 10)**	The manual does not describe loading-tube removal — ask the manufacturer.	▶ 2:00
 3D	**⑤⑥ Main diaphragm assembly · lever** Tip the assembly so the lever (26) slips out of the pusher post, lift it out, then inspect the diaphragm parts and lever (p. 9).	—	Replace damaged parts (kit R299X000012).	▶ 2:26
 3D	**⑦⑧ Supply tubing · lower casing off the body** Disconnect the pilot supply tubing and remove the 2 cap screws (18) to separate the lower casing from the body. The body stays in the line (p. 10).	Tubing 18 mm · cap screws 3/8 in. hex key **Torque: check the manual value**	Inspect the body O-ring (15) and replace if needed (p. 10).	▶ 2:53
 3D	**⑨ Disk · orifice** Unscrew the disk holder from the stem and check the disk and orifice for nicks or damage (p. 10).	Orifice: 41 mm socket **Orifice: 136–163 N·m, lubrication per the manual (p. 10)**	If the orifice size changes, update the nameplate (p. 10).	▶ 3:22
 Real 299HS 2006	**⑩–⑫ Pilot inlet fitting · disk · check valve** Remove the 4 screws (46) and the inlet fitting, then inspect the pilot disk, orifice, check valve (45) and bleed (70) (p. 11).	Screws 10 mm · disk 14 mm · check valve 16 mm · pilot orifice 13 mm socket **Screws (46): 3.4–4.5 N·m (p. 11)**	Lightly lubricate the O-rings (49, 54) (p. 11) · if clogging is suspected, remove the elbow (19) and clean the inlet screen (51) (p. 11).	▶ 3:36
 Real 299HS 2006	**⑬ Bonnet · control spring** Unscrew the bonnet (34) with the adjusting screw and remove the spring seat and control spring (p. 11).	Bonnet: 375 mm motor wrench	Spring colour = set range — check Table 2 and the nameplate.	▶ 4:24
 Real 299HS 2006	**⑭–⑯ Pilot spring case · diaphragm** Remove the 8 screws (30) and the spring case, then tip the diaphragm assembly off the lever (p. 11).	Screws: Phillips screwdriver **Screws (30): 1.4–2.0 N·m, crisscross (p. 11)**	On reassembly, point the vent (27) down — prevents clogging and moisture (p. 11).	▶ 4:39
 3D	**⑰⑱ 67CF diaphragm · filter** Isolate and depressurize the 67CF. Back out the adjusting screw (18) to remove spring compression before removing the 6 spring-case screws. Inspect the diaphragm; remove the 4 dripwell screws to inspect the filter and cartridge (67C pp. 5–7).	**Spring-case and dripwell screws: 1.7–3.4 N·m; case screws crisscross (67C pp. 6–7)**	Replace damaged parts — the filter can be cleaned if no spare is at hand (67C p. 6).	▶ 5:12
 3D	**Assembly · checks · start-up** Assemble in reverse — set 13 mm from the top of the upper-casing opening to the top of the spring seat (p. 10), check tubing, wire seal and tag, then set the pressures during start-up.	—	★ Do not assume the previous settings hold after maintenance — reset them during start-up (p. 11).	▶ 6:11

Photos: Real = KFNGS service records (year · 299HS = parts shared with the 299H) · 3D = KFNGS guide screen (training reconstruction) · ①… = guide video steps

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Installation, operation and maintenance must be performed by qualified personnel following applicable codes, site safety procedures and the manufacturer's manual.
The disassembly & assembly guides are training reconstructions, not manufacturer procedures. Do not perform any work the manual does not describe, including releasing spring force, until you have checked the manufacturer's procedure.
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